AI System to Detect Mental Health Signals from Gaming & Social Media Behavior

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INTRODUCTION

Mental health is a growing concern with rising use of social media and gaming. Many individuals are unaware how digital habits affect daily mood and stress levels.

This project applies machine learning to analyze self-reported behavioral patterns from a survey of UNC Charlotte students and the Charlotte, NC community to detect potential mental health signals.

RESEARCH QUESTIONS

- How does social media & gaming usage affect daily mental health?
- Does increased screen time correlate with worse mood?
- Does gaming reduce or increase stress levels?
- Can AI predict mental health signals from user behavior?

DATASET & PREPROCESSING

Source Self-collected Google Form, UNC Charlotte community, April 2026

Size 100 raw responses → 70 usable rows, 11 features after cleaning

Target Daily Mood — Bad (0–1) | Neutral (2) | Good (3–4)

Split 80/20 train/test → 56 training, 14 test samples

PREPROCESSING STEPS

1. Dropped 17 irrelevant / free-text columns
2. Ordinal encoding for Social Media & Video Game Hours
3. Median imputation (SimpleImputer) for missing values
4. Target binned: Bad (0–1), Neutral (2), Good (3–4)
5. 80/20 train/test split → 56 train, 14 test samples
6. Removed erroneous 'Yes' value from Social Media Hours

DATA PIPELINE

Survey → Clean & Encode → EDA → Train Model → Evaluate

Key Features: Social Media Hours, Video Game Hours, Anxiety Levels, Social Comparison, Self Esteem, Screen Use for Relief, Loneliness, Sleep Affected, Addiction Level, Gamer Status.

METHODS & MODELS

SVM
Support Vector Machine — RBF kernel, class_weight='balanced'. Finds max-margin hyperplane: minimizes $||w||^2/2$ s.t. $y_i(w \cdot x_i+b) \geq 1$.

KNN (k=6)
K-Nearest Neighbors — Euclidean distance, StandardScaler. k=1–20 tuned; k=6 optimal with best train-test gap (58.9%→64.3%).

Linear Regression
Baseline: tests continuous mood prediction. Ordinal/categorical target makes this unsuitable → $R^2=-0.24$ confirms non-linear relationship.

Calibrated SVM
CalibratedClassifierCV wraps SVM via isotonic regression (5-fold CV), producing reliable probability scores per mood class.

Decision Tree (Feature Selection)
DT (max_depth=4) trained on ALL features vs. screen-time ONLY. Gini impurity ranks importance: psychological vars outrank screen hours.

RESULTS

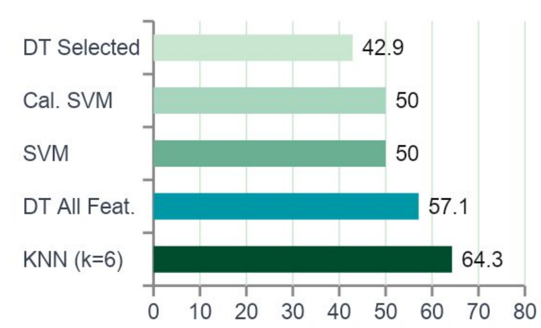
64.3% KNN Best Test Acc.

57.1% Decision Tree (All)

50.0% SVM / Cal. SVM

−0.24 Linear Reg. R^2

Model Accuracy Comparison (Test Set %)

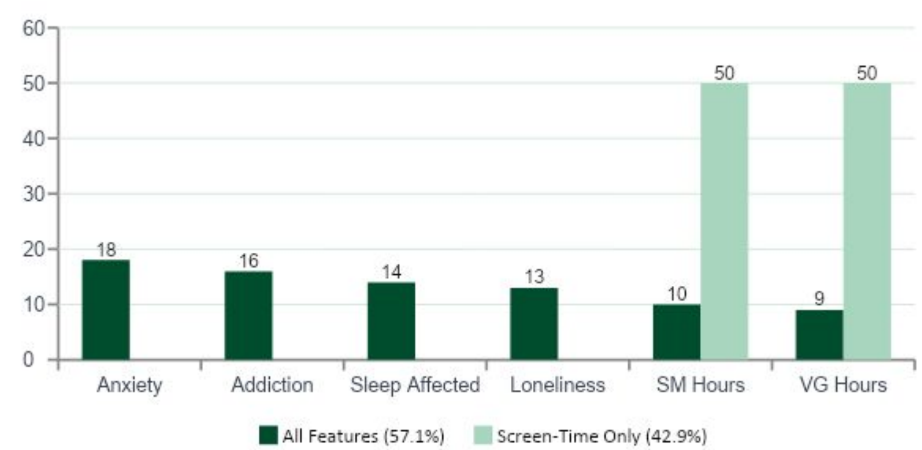


KNN Classification Report (k=6):

Class	Precision	Recall	F1
Bad	0.00	0.00	0.00
Neutral	1.00	0.33	0.50
Good	0.58	1.00	0.74
Overall			**64.3 %**

💡 **Key Finding:** Psychological side-effects (anxiety, addiction, sleep disruption, loneliness) ranked higher in feature importance than raw screen-time hours. Screen-time-only model accuracy dropped 57.1% → 42.9%.

FEATURE IMPORTANCE: ALL FEATURES VS. SCREEN-TIME ONLY



KEY FINDINGS

KNN (k=6) Was Best Model
64.3% test accuracy; smallest train-test gap (58.9% train) — indicating least overfitting among all tested models.

Psychological Factors Beat Screen Time
Anxiety, sleep disruption, addiction & loneliness outranked raw screen-time hours as predictors of daily mood.

Avoidance Coping Backfires
Respondents using screens to escape stress reported lower moods — consistent with maladaptive coping theory in psychology.

Mood Is Non-Linear
Linear Regression $R^2=-0.24$ confirms mood cannot be predicted linearly from screen time, validating classifier approach.

Calibration Adds Interpretability
Probability Calibration produced reliable confidence: Bad 3%, Good 54%, Neutral 44% — useful for real-world transparency.

LIMITATIONS & ETHICS

Small Sample: ~70 usable rows, mostly UNCC students. Results may not generalize nationally.

Self-Reported Bias: Respondents may misremember screen time or mood at time of survey.

Class Imbalance: Only 1 'Bad' sample in test set; all models failed to predict this class.

Correlation ≠ Causation: Academic stress may independently explain both screen time and low mood.

Privacy: All responses fully anonymous; no personally identifiable information collected.

Not Clinical: For awareness and education only — never use for diagnosis or screening.

REFERENCES

- Psychology Today — Benefits of social media for community connection.
- Engadget — 9% of teens negatively impacted; 37% sleep affected by TikTok.
- Open Textbooks — Cognitive benefits of video games.
- Pew Research Center — Teen pressure and social media anxiety.
- Scikit-learn — CalibratedClassifierCV, KNN, SVC, DecisionTree.

ACKNOWLEDGEMENT

Survey distributed to UNC Charlotte students & Charlotte, NC community, April 2026. All participation was voluntary and anonymous. Thanks to the DTSC instructional team for guidance throughout this project.

AI Transparency: Claude (Anthropic, claude-sonnet-4-6) was used to help fix code errors.